

MATSim for Jakarta MRT Expansion: Projected Ridership and Mode Shifts

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Abstract

This study extends a large-scale MATSim model of Greater Jakarta to evaluate the proposed expansion of the Jakarta Mass Rapid Transit (MRT) system. The inherited scenario represents approximately three million agents, corresponding to a 10% sample of the metropolitan population, together with multimodal networks and full-day activity-travel plans. MRT is introduced as a distinct mode with its own alternative-specific constant and coefficients for in-vehicle time, waiting time, and station access-egress time. Automated Fare Collection smart-card records were processed into observed passenger flows and used to calibrate the existing 13-station scenario before adding seven proposed stations. The calibrated scenario generated 12,049 daily MRT trips, compared with an observed-equivalent target of approximately 11,140 trips. Expansion to 20 stations increased simulated MRT use to 15,284 trips, a rise of 26.85%. Trip-level matching identified 3,841 journeys switching from non-MRT modes, dominated by motorcycles (76.65%), followed by cars (15.10%) and conventional public transport (8.25%). The application illustrates how MATSim can support prospective rail appraisal while revealing the modal origin and traveller composition of additional demand.

Keywords: MATSim; agent-based modelling; MRT expansion; smart-card data; mode substitution; Greater Jakarta

1. Motivation

Greater Jakarta is a large and increasingly polycentric metropolitan region characterised by suburban growth, urban sprawl, and intensive commuting between peripheral municipalities and major activity centres (Pratama et al., 2022; Winarso et al., 2015). Within this dispersed urban structure, motorcycles remain highly competitive because they provide inexpensive, flexible, and door-to-door mobility under congested conditions and uneven public transport coverage (Ilahi et al., 2021). However, high motorcycle dependence also contributes to congestion, safety risks, and environmental burdens.

Jakarta MRT Phase 1 currently operates along a 13-station north–south corridor. Its proposed northward extension provides an opportunity to examine whether improved rail coverage can attract additional passengers and reduce dependence on road-based modes before the infrastructure is completed. From a transit-oriented development perspective, the effectiveness of rail investment depends not only on infrastructure provision but also on station accessibility, surrounding land use, connectivity, and pedestrian conditions (Cervero & Kockelman, 1997; Ibraeva et al., 2020). Network expansion may therefore alter station catchment areas, access and egress conditions, transfer opportunities, and the relative attractiveness of MRT compared with competing modes.

Three research gaps motivate this study. First, many TOD assessments remain static and place-based, focusing on indicators such as density, land-use diversity, accessibility, walkability, and station typologies (Abdullah et al., 2024; Papa & Bertolini, 2015). Although useful for evaluating station-area conditions, these approaches do not simulate how individual travellers adapt their daily activity–travel plans when the rail network changes. Second, conventional ridership studies often estimate total passenger growth without identifying the previous modes of new rail users. This limits policy interpretation because a shift from motorcycle travel has different implications from a shift within the public transport system. Third, relatively few studies have evaluated rail expansion in a motorcycle-dependent developing metropolis using an agent-based model calibrated with observed smart-card demand.

Agent-based modelling is suitable for addressing these limitations because it represents individual activity schedules, travel choices, and interactions with the transport network. MATSim enables agents to adapt their modes, routes, and departure times through iterative plan execution, scoring, and replanning (Horni et al., 2016). This study extends the existing Greater Jakarta MATSim framework developed by Ilahi & Axhausen (2019) and (Ilahi et al., 2021) by representing MRT as a distinct travel mode and calibrating the existing 13-station network using Automated Fare Collection origin–destination data. The calibrated model is then expanded to 20 stations to evaluate changes in MRT ridership, the modal origins of new MRT trips, activity purposes, station demand, and traveller profiles. By tracing individual trips

across scenarios, the study provides behavioural insights that cannot be obtained from aggregate ridership forecasting alone.

2. MATSim scenario extension and calibration

This study extends the Greater Jakarta agent-based transport model developed by Ilahi et al. (2026), as illustrated in Figure 1. The inherited model represents approximately three million agents, equivalent to 10% of the metropolitan population, and includes an OpenStreetMap- and GTFS-based transport supply, a census-based synthetic population, and daily activity–travel demand. These components form Scenario 1 (S1), in which MRT is included within general public transport but is not represented as a distinct mode. Scenario 2 (S2) introduces the existing 13-station MRT network as a separate mode and calibrates MRT-specific parameters using 2024 JakLingko smart-card origin–destination data. Scenario 3 (S3) adds seven proposed stations, creating a 20-station network. Population, non-MRT transport supply, travel demand, and calibrated behavioural parameters remain unchanged between S2 and S3, allowing the effects of network expansion on ridership, mode shifts, activity purposes, and traveller profiles to be isolated.

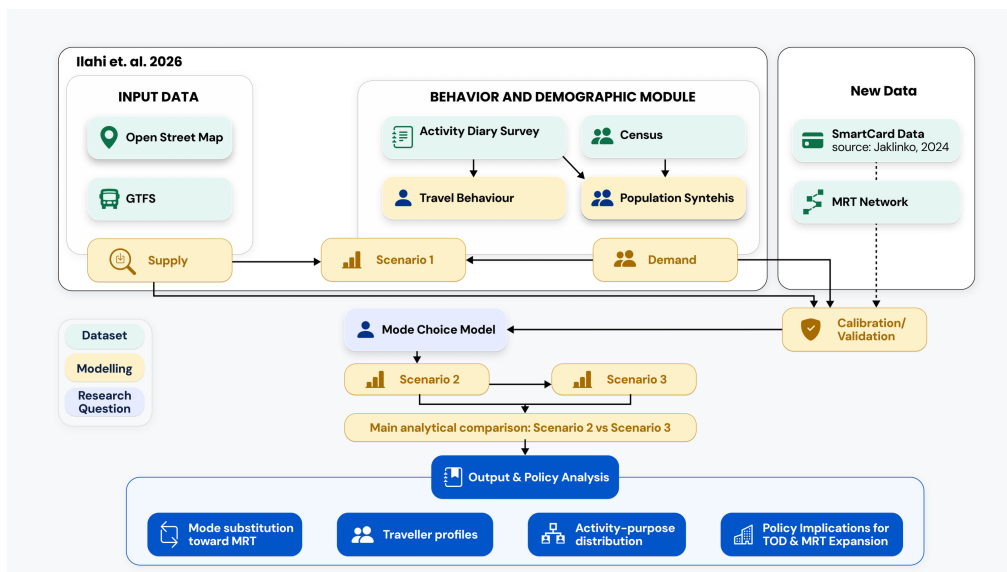


Figure 1. Development, calibration, and analytical framework of the Greater Jakarta MATSim scenarios.

3. Results

The calibrated Scenario 2 generated 12,049 daily MRT trips at the 10% simulation scale, compared with an observed equivalent of 11,139 trips from the 2024 JakLingko smart-card data. The relative difference of 8.16% indicates that the model reproduced the overall magnitude of existing MRT demand sufficiently for scenario analysis. Expanding the MRT network from 13 to 20 stations increased daily MRT trips from 12,049 in S2 to 15,284 in S3, corresponding to an increase of 3,235 trips or 26.85% (Table 1). Because both scenarios use the same 10% synthetic-population sample, this percentage directly represents the simulated effect of the network expansion.

Table 1. Simulated MRT ridership and mode-shift records under the existing and expanded networks

Mode	Trip		Change
	Scenario 2	Scenario 3	
MRT	12049	15284	3235
Motorcycle	2944	434	-2510
Car	580	95	-485
Public transport (PT)	317	30	-287

Trip-level matching identified 3,841 journeys that changed from non-MRT modes in S2 to MRT in S3. Motorcycles were the dominant source, accounting for 2,944 trips or 76.65% of all matched shifts, followed by cars at 15.10% and conventional public transport at 8.25%. Home- and work-related travel

represented more than 90% of MRT use, while the largest user increases occurred among travellers aged 26–45 years. These results indicate that the expansion mainly strengthens regular commuting and improves MRT competitiveness against motorcycle travel. The findings highlight the importance of combining rail expansion with reliable feeder services, improved pedestrian access, integrated fares, coordinated transfers, and appropriate motorcycle park-and-ride facilities. Such measures are particularly important around the proposed northern stations, where first- and last-mile access conditions may determine whether the projected ridership growth can be realised.

4. Discussion and outlook

The application demonstrates a practical pathway for extending an established MATSim scenario when a rail mode was previously embedded within generic public transport. Three aspects are especially relevant to the MATSim community. First, the explicit MRT utility allows rail-specific access, waiting, and in-vehicle conditions to influence daily plan evaluation. Second, observed smart-card demand provides an empirical anchor before a prospective infrastructure experiment is conducted. Third, trip-level comparison across otherwise identical scenarios makes it possible to identify the transport modes supplying additional MRT demand, rather than reporting only net ridership change.

The dominance of motorcycle-to-MRT substitution is particularly important for metropolitan regions where two-wheelers are both competitors to mass transit and potential feeder modes. The result implies that rail expansion should be assessed together with station access, walking conditions, feeder services, fare integration, and motorcycle park-and-ride arrangements. These complementary interventions can be introduced as subsequent MATSim scenarios to distinguish the effect of network extension from the effect of improved first- and last-mile integration.

The present results should be interpreted in light of the 10% synthetic-population sample and the remaining uncertainty in the spatial representation of station catchments. Future work will strengthen station-level validation through spatial weighting, test alternative feeder and fare policies, and examine the robustness of results across population samples and parameter settings. Nevertheless, the current model provides a calibrated platform for evaluating future Jakarta MRT investments before implementation and for tracing behavioural responses that are difficult to recover from aggregate forecasting alone.

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